

Experimental Review on Bulk Properties and Light/Strange Production in Heavy-Ion Collisions

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arXiv:2607.11640v1

A review of experimental results presented at SQM2026

Main motivation

- Understand the bulk properties of the medium produced in collisions
- Understand where does partonic behaviour emerges and disappears

Bulk properties

- Equation of State (EoS), radial flow and fluctuations, bulk and shear viscosity, collective/partonic flow, etc.
- Speed of sound, NCQ scaling, Ω/ϕ ratio

Speed of Sound from Mean Transverse Momentum



Caleb Broodo (University of Houston) for the STAR Collaboration

Extracting the speed of sound from
 $\langle p_T \rangle$ measurements in Au+Au
collisions at RHIC BES-II program

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The 22nd International Conference on
Strangeness in Quark Matter
22-27 March, 2026, Los Angeles, CA



Speed of Sound from Mean Transverse Momentum

Definition:

$$c_s^2 = \frac{dP}{d\epsilon} = \frac{d \ln T}{d \ln s}$$

How the Pressure responds to a change in Energy Density.

- Makes it directly sensitive to the equation of state (EoS).

Across the phase transition, LQCD predicts a temperature dependence.

Measuring c_s^2 allows us to experimentally test that EoS.

We can't stick a thermometer into a detector...

Speed of Sound from Mean Transverse Momentum

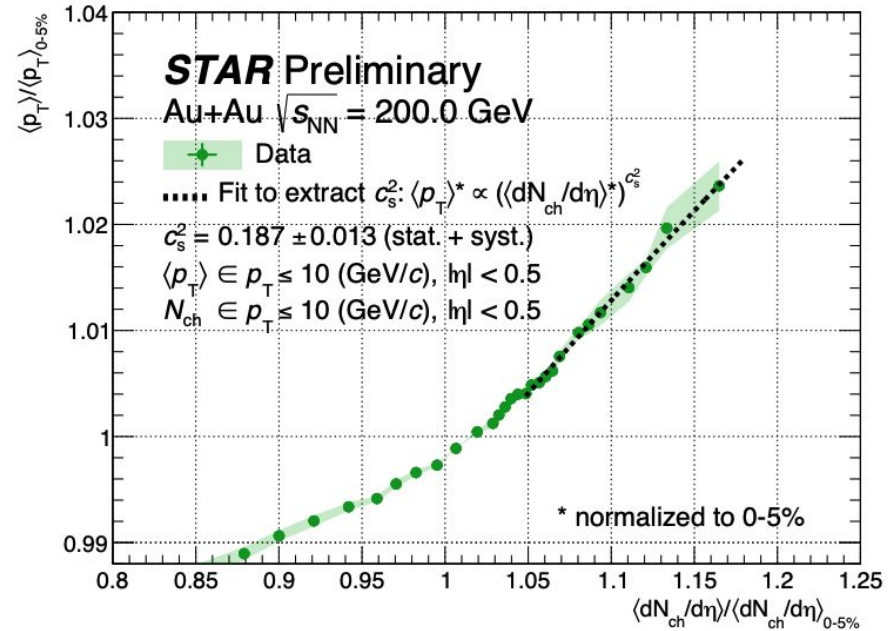
Use $\langle p_T \rangle$ as a proxy for T

Use N_{ch} as a proxy for entropy s

- Possible by restricting $b \sim 0$
- Geometry approx. fixed
- Fluctuates in Entropy, not V

$$c_s^2 = \frac{d \ln T}{d \ln s} \Rightarrow c_s^2 \approx \frac{d \ln \langle p_T \rangle}{d \ln N_{ch}}$$

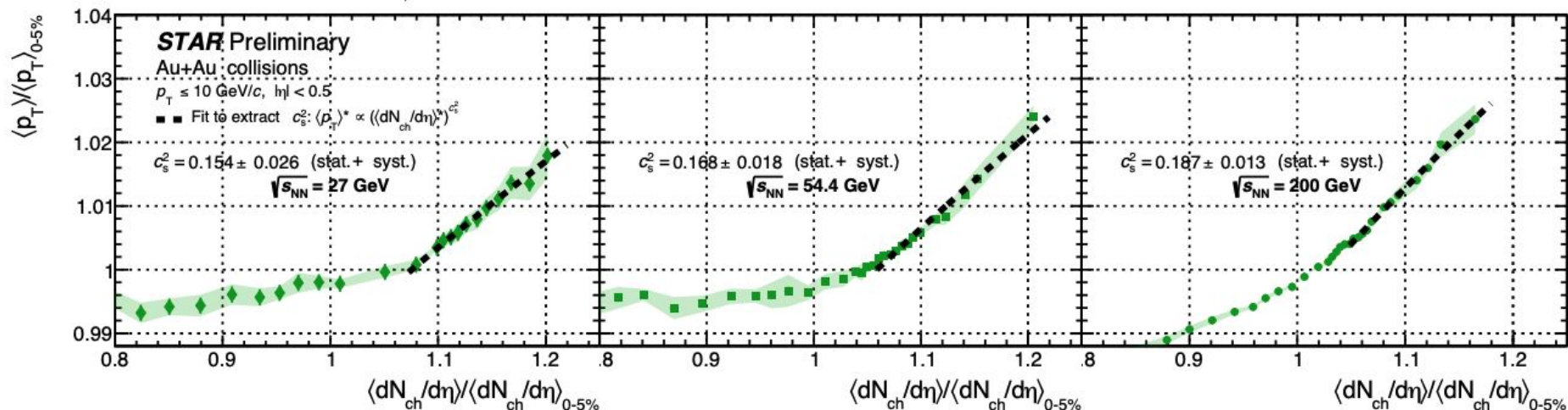
Extracting c_s^2 from the $\langle p_T \rangle$ response to multiplicity fluctuations.



Speed of Sound from Mean Transverse Momentum

Energy dependence:

c_s^2 increases with $\sqrt{s_{NN}}$ →



Systematic uncertainties are determined by varying the $\langle p_T \rangle$ range on the fit used to extrapolate $p_T \rightarrow 0$

Speed of Sound from Mean Transverse Momentum

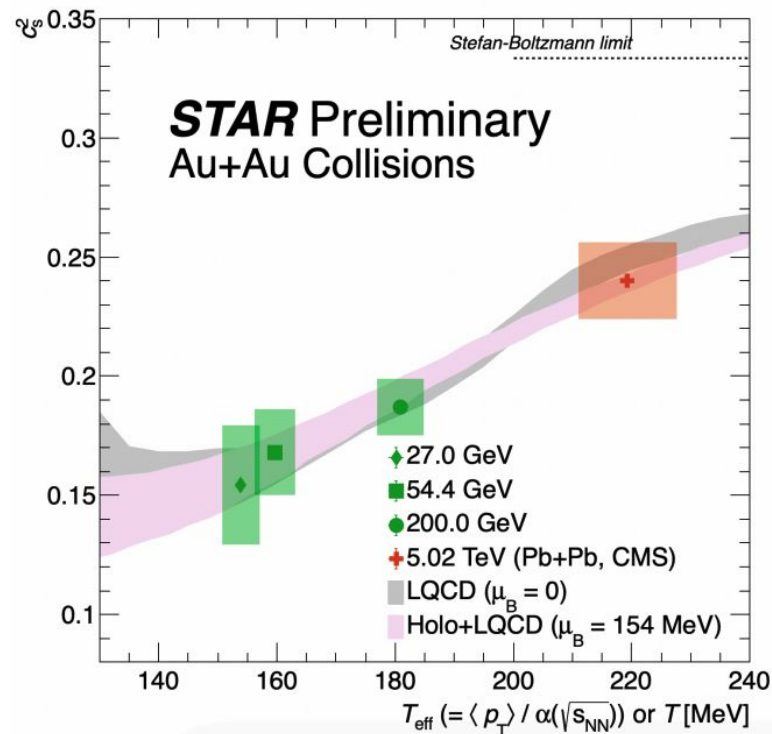
After extracting across the BES-II

- experimental T dependence of c_s^2 can be built, as shown here.

Plotted against an effective Temperature

The experimental T dependence agrees reasonably with LQCD.

- Points sit within LQCD predictions
- Follow the increasing trend



Number of Constituent Quarks Scaling

NCQ “Number of Constituent Quarks” scaling is used as a probe for partonic collectivity.

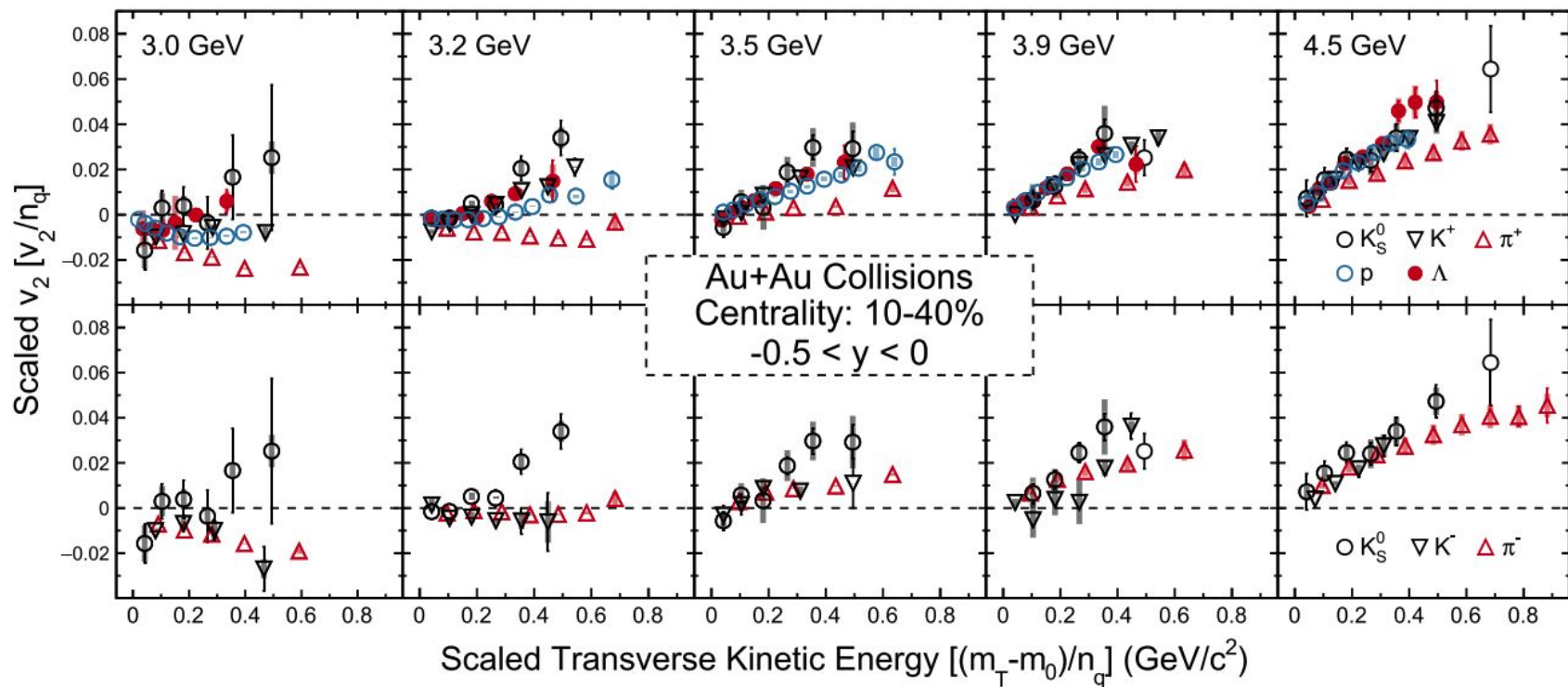
In the coalescence view, the flow is can roughly be described as:

$$v_2(\text{hadron}) \approx n_q \cdot v_2(\text{quark})$$

$$v_2/n_q \quad \text{vs.} \quad (m_t - m_0)/n_q$$

After scaling by n_q , a common curve found across species suggests collective behaviour before hadronization.

Number of Constituent Quarks (NCQ)



Number of Constituent Quarks Scaling

At lower beam energies, the species follow their own behavior

As you increase collision energy, the species begin to converge.

Somewhere around 3.9 GeV, partonic interaction are becoming dominant

- NOT a phase boundary!
- Evidence of the onset of collectivity as the beam energy increases

Strangeness Production using Ω/ϕ



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Exploring strangeness production across Beam Energies in Au+Au Collisions at STAR

Weiguang Yuan, Xiongiong Xu (for the STAR Collaboration)
Tsinghua University

- Motivation
- STAR detector and strange hadron reconstruction
- Results
 - ✓ p_T and rapidity spectra
 - ✓ Energy dependence of R_{CP} and $\bar{B}/B$
 - ✓ Ω/ϕ ratios
- Summary

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Strangeness Production using Ω/ϕ

$$\Omega^- = sss \quad \phi = s\bar{s}$$

Both probe strange production, and we can see the baryon/meson ratios. These ratios are indicative of the production mechanism

- Enhanced baryon production is consistent with a partonic-coalescence picture

Are they seeing the ratio enhanced in central collisions, and can a partonic-coalescence model explain it?

Strangeness Production using Ω/ϕ

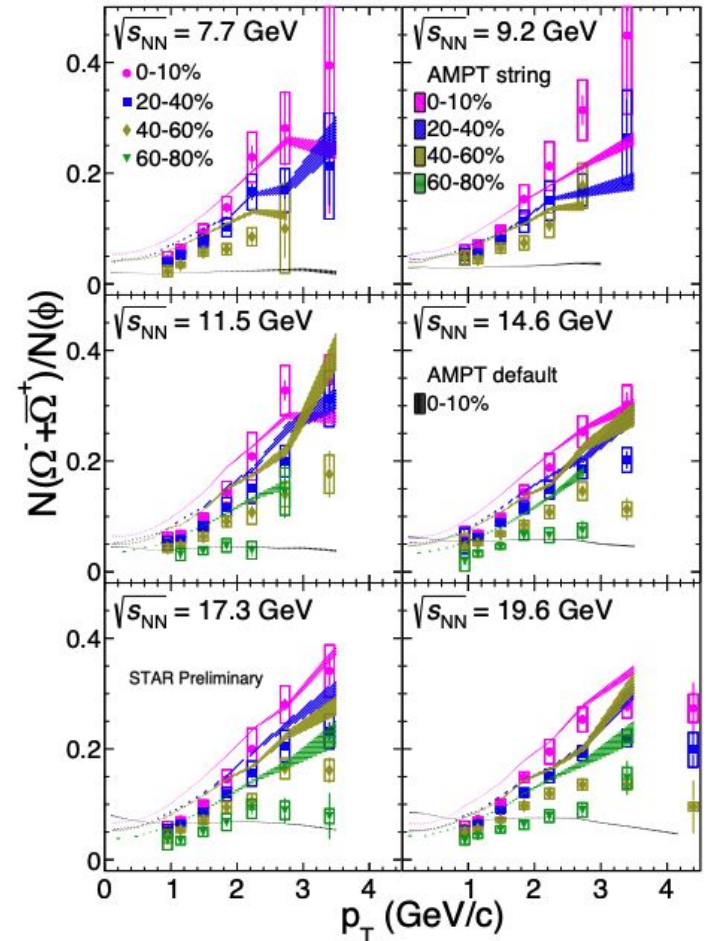
Ω/ϕ vs. p_T

Colors are different centralities, and is shown across various $\sqrt{s_{NN}}$.

Tried AMPT and String-Melting AMPT

- String-melting was successful at describing the data
- String-Melting model includes partonic DoF and coalescence

These data are consistent with strange-quark coalescence in Central Au+Au collision.



Conclusion / Summary

- STAR's extraction of the Speed of Sound gives experimental access to the QCD EoS and agrees well with QCD
- Collectivity appears with increasing collision energy
 - NCQ scaling is broken at lower beam energies, and gradually appears around $\sqrt{s_{nn}} \sim 4.5\text{GeV}$
- Strange hadrons from data show evidence of deconfinement consistent with the string-melting AMPT model (at least down to 7.7 GeV).

Results are moving away from “Does QGP form” to describing its properties, and where it's regime approximately exists.

References

S. Shi and X. Zhu, *Experimental Review on Bulk Properties and Light/Strange Hadron Production in Heavy-Ion Collisions*, arXiv:2607.11640v1.

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STAR Collaboration, *Onset of Constituent Quark Number Scaling in Heavy-Ion Collisions at RHIC*, Phys. Rev. Lett. 135, 072301 (2025).

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